Unsupervised Learning and Dimensionality Reduction

K-Means and EM Clustering on Reduces Feature Spaces with PCA, ICA, amd RP

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***Abstract*—This paper evaluates an unsupervised pipeline that discovers hidden dataset structures using K-means and Expectation Maximization (GMM), and subsequently applies Principle Component Analysis(PCA), Independent Component Analysis (ICA), and Randomized Projections(RP) to reduce unnecessary dimensionality. The constructed low-dimensionality representations are passed into a pre-tuned neural network to demonstrate how the additive clustering components and meaningful feature reductions guide the predictions along the hyperplane. We see that results from datasets with dominating features achieve better accuracy using hard, discriminatory cluster assignments and variance-focused dimensionality reduction. It also shows how noisy sets with even feature variance demonstrate improvement from probabilistic assignments, especially after top influential attributes are exposed with ICA [1].**

INTRODUCTION

Objective

It is easy to assume that poor predictions stem from inadequate data. Supplying more data appears to be an intuitively reasonable way to further generalizations and enhance discriminative patterns within the feature space. While true under some conditions, increasing dimensionality eventually offers diminishing returns, often replacing predictive accuracy with added complexity. This is the curse of dimensionality [2]. This paper aims to challenge a common misconception and mitigate this statistical curse through an exploration of clustering and dimensionality reduction algorithms that highlight informative features, eliminate redundant dimensions, and assess algorithmic speed, complexity, and performance.

Hypotheses

Global: With the understanding of scaling complexity as a universal machine learning problem, we will attempt to improve model prediction efficiency by enhancing meaningful signals while muting irrelevant ones. Reducing irrelevant features should exhibit clearer predictive results and avoid any unrelated noise. If sufficient dimensionality reduction is achieved and meaningful components are extracted through clustering, then both datasets, despite inherent limitations, should yield improved performance compared to identical models trained on the original, unprocessed datasets.

Clustering: Due to significant target imbalance present in the Hotels set, the results will likely suffer from soft probability assignments from EM, whereas K-Means is expected to demonstrate improved accuracy with hard target class predictions. Soft assignments may weigh larger classes more heavily due to their inherently larger spread and dispersion, encouraging sharper assignment splits among outliers and differentiating them from smaller classes [3]. In contrast, analysis of Accidents should benefit from soft decisions, introducing greater flexibility across the noisy feature space, which should smooth out cluster overlap.

Dimensionality Reduction: ICA is expected to excel on the Hotels set by effectively isolating statistically independent factors and pulling apart driving signals. Because noise in the Hotels dataset is likely generated by correlated features, distinguishing feature "voices" will provide a clearer distinction between target classes and allow for more distinct class predictions [1]. As for the larger dataset, PCA should outperform ICA and RP, as its focus on achieving high variance will determine the few features that shoulder much of the class division, with an additive effect of reducing irrelevant signals from noisy attributes.

Neural Networks: Dimensionality reduction mitigates noise within a dataset while preserving variance, enabling the neural network to converge more efficiently using fewer attributes. If the data transformation introduces no additional structure but exclusively removes information, then the network may display higher loss scores in exchange for faster computation. But the additive benefit of cluster-derived feature sets should improve metric performance while simultaneously maintaining low time complexity [4]. The two techniques should effectively strip out irrelevant features, substituting them with ones that reveal new, informative signals.

Preceding Research

Previous work presents optimally trained hyperparameters tailored to each dataset for the neural network used in this study. Minor adjustments were applied to the parameters and training procedure for improved output and enhanced performance [5].

DATASET METHODOLOGY

Hotel Booking Demand

The first dataset to be tested, Hotel Booking Demand (H1/H2) [6], [7], is medium sized, with about 119,000 rows and measuring 32 features relating to client reservation information. Cleaning the set, such as removing leakage, optimizing date time values, and dropping features that produced little to no information gain, resulted in a size reduction of roughly 95% and a total of 87,138 rows across 14 features. Of that, a 80/20 hold-out split of 69,710 training and 17,428 testing examples was implemented. 'is canceled', a binary feature determining whether the client canceled their reservation, was chosen as the target attribute, splitting the set about 27% to 73% between true and false, respectively. The heavily imbalanced data within Hotels introduces a unique challenge for binary classification predictions, as much of the set being classified to one group may create a lack of sufficient training examples and drown out the minority class.

US Accidents

A larger, more practical set, US Accidents (since 2016) [8], contains over 8 million rows and 46 columns. Due to its size, a stratified subset of approximately 1.2 million pre-cleaned rows was used for this report. It contains car accident crash data across the United States, including location, time, and weather. The dataset was cleaned similarly to the Hotels set, with many features dropped due to their minuscule information gain. To help cut down memory costs, several features were converted to smaller datatypes, truncated, or summarized into categorical values. The preprocessing resulted in 28 features across 1,196,125 rows. The same 80/20 split yields 717,675 rows to train and 239,225 to test. But when training a neural network on this dataset, a 60/20/20 train/validation/test set was used, leaving a total of 478,450 rows to train on. The network implemented for this set will perform regression testing to predict the derived length between Start Time and End Time (Duration Seconds). The Accidents dataset, given its size, presents time complexity as an obstacle that bottlenecks results and adds additional noise, making it difficult for clustering algorithms to split examples into clearly defined groups.

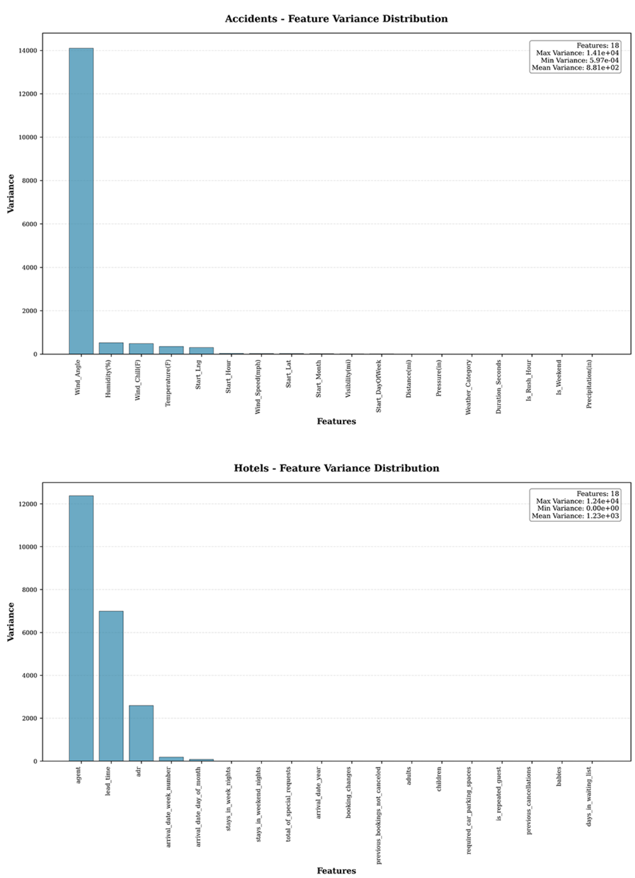

Fig. 1. Variance graphed before DR applied (Top: Accidents; Bottom: Hotels)

Preprocessing

The following was performed on both datasets to lessen skewed results and reduce memory expenses. Winsorizing was performed to minimize the top and bottom one percent of the dataset. Using scikit-learn [9], each value was standardized between 0 and 1 with StandardScaler, while missing values were either filled or dropped using SimpleImputer. Features holding integers and floating-point numbers were classified as numeric, while the number of unique values separated low categorical from high categorical features. CountEncoder was used for categorizing the data to substantially lessen memory usage and dimensionality. Features with low to high cardinality were encoded using their frequencies. All transformations were orchestrated in a pipeline via ColumnTransformer. Finally, the processed data is wrapped into three Dataloaders for efficient storage and retrieval.

TABLE I
DATASET SUMMARY STATISTICS

Dataset	Structure	Cleaned	ML Problem	Target Dist.
Hotels	29 : 119,390	14 : 87,138	Classification	0: 63,371 (72.7%) 1: 23,767 (27.3%)
Accidents	40 : 1,200,000	28 : 1,196,125	Regression	mean 8.398 std 0.837

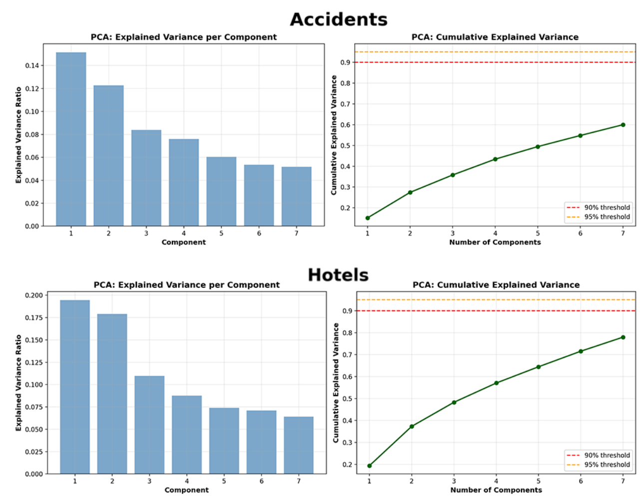

Fig. 2. Variance graphed after DR applied (Top: Accidents; Bottom: Hotels; Left column: sorted component variance bar graph; right column: Cumulative variance explained curve)

ALGORITHM IMPLEMENTATION

This layered study comprises three primary Python modules, encompassing an abstract implementation of clustering, dimensionality reduction, and neural network prediction. Each module defines a base class to manage primary experimental data for its respective models, leveraging parent-child inheritance to minimize code redundancy and streamline development. Shared utility functions for data processing, cache management, logging, and plotting further reduce duplicate code. Isolated core functional logic enables convenient experiment adjustments.

Clustering

Clustering is performed without data labels to reveal raw feature patterns, using a scaled Euclidean distance metric to prevent feature dominance. Cluster counts ranging inclusively from two and eight were evaluated to balance coverage and computational efficiency. Each model executed ten random initializations per clustering run to enhance stability and convergence, followed by five additional runs at the optimal cluster count to assess its stability and consistency across different seeds.

The optimal cluster configuration was determined using a minimum rank sum across multiple cross-algorithmic evaluation measurements: Silhouette Score, Dunn Index, Davies-Bouldin Index(DBI), and Calinski-Harabasz Score(CBI). Each metric ranked candidate cluster values from best to worst, and the candidate with the lowest composite rank was chosen. This approach mitigates metric-specific bias by equally weighing cluster cohesion, separation, similarity, and dispersion. Akaike Information Criterion(AIC) and Bayesian Information Criterion(BIC) were excluded, as their incompatibility with K-Means would distort comparisons with GMM results.

This study implements the scikit-learn `Kmeans` and `GaussianMixture` classes.

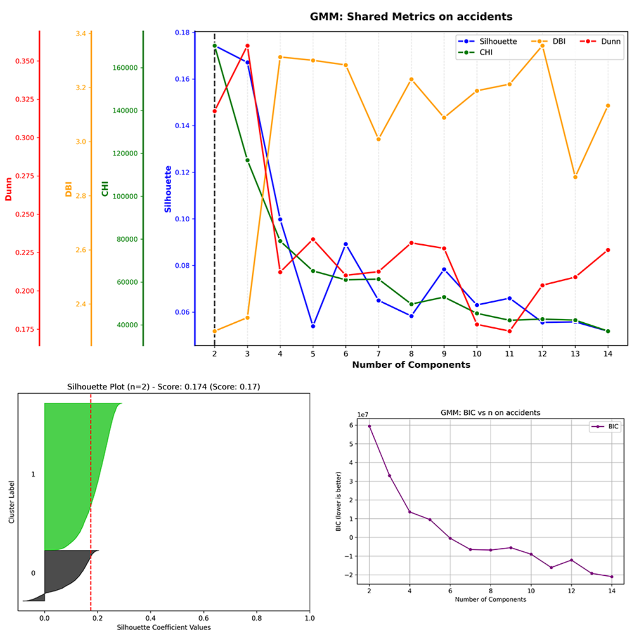

Fig. 3. GMM clustering metrics on Accidents before dimensionality reduction (Top: Dunn, DBI, CHI, Silhouette; Left: Silhouette plot; Right: BIC curve)

Fig. 4. Clustering Quality Metrics Heatmap on Accidents

K-Means: K-Means uses hard clustering, assigning each data point to exactly one centroid-based cluster. Although not directly useful in this study, inertia was calculated to measure overall cluster compactness.

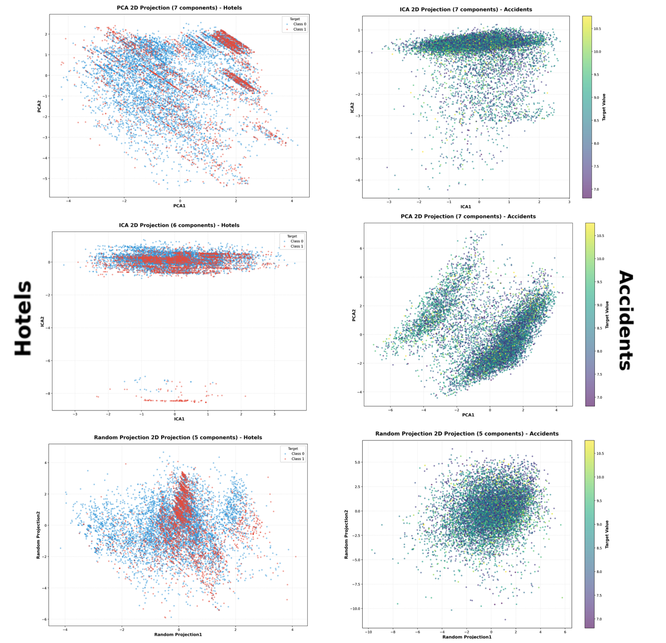

Fig. 5. 2d representation scatter plot using components with the highest variance after DR applied (Left column: Hotels; Right column: Accidents; Top: PCA; Middle: ICA; Bottom: RP)

Gaussian Mixture Model: The Gaussian Mixture Model applies Estimation Maximization to determine the number and composition of Gaussian components. It performs soft, distribution-based clustering, assigning each instance probabilistic likelihoods of belonging to each cluster. Cluster visualizations are plotted with probabilistic means. Convergence tolerance was set to 0.001 and iterations were capped at 100 for both datasets to efficiently scale complexity. For the Hotels dataset, covariance types were set to "full" to capture unconstrained feature correlation, while "diagonal" covariance was used for the Accidents dataset to reduce overfitting the excess noise and minimize computation costs. AIC, BIC, and log likelihood were also recorded.

Dimensionality Reduction

The following algorithms are used to transform high-dimensional data into lower-dimensional representations while preserving the structure and patterns of the original data.

Principle Component Analysis: Focusing the models on meaningful structures that effectively separate the data enables efficient clustering downstream and subsequent neural network training. Component counts between two and eight were tested and measured. The `FastICA`, `PCA`, and `GaussianRandomProjection` scikit-learn classes were used for these transformations.

Independent Component Analysis: FastICA is implemented to maximize statistical independence by distinguishing mixed signals from correlated features and dividing them into separate, independent sources. By identifying non-Gaussian features, it strives to find the minimum components that achieve 85 percent of the maximum kurtosis. Both datasets were whitened to unit variance to normalize the data and stabilize the multivariate signals. The maximum iterations were set to 200 with a tolerance of 0.0001, as the algorithm efficiently converged. Reconstruction is approximated via the unmixing matrix, where MSE measures residual dependence.

Randomized Projection: Randomized Projections apply random Gaussian metrics to preserve 95 percent of its variance and pairwise distances while projecting data onto lower dimensions. Rather than focusing on variance or independence, RP employs random linear transformations, prioritizes speed with simple calculations, and manages scaling data with inexpensive tasks. This algorithm is evaluated by its reconstruction error and pairwise distance preservation.

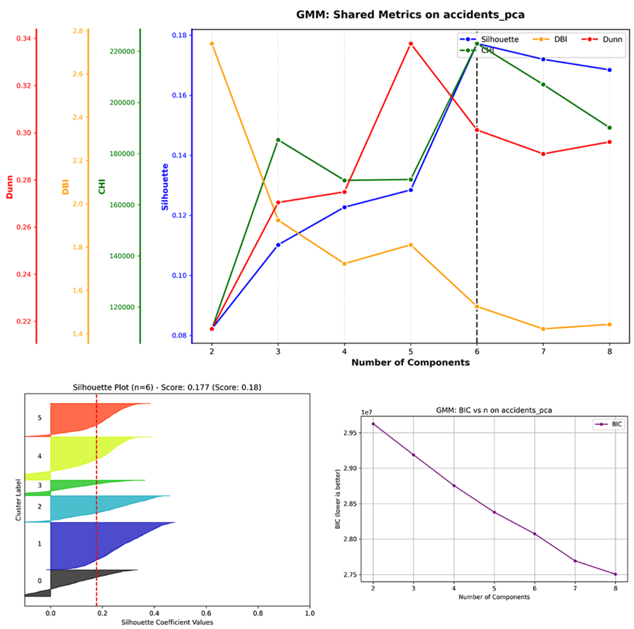

Fig. 6. GMM clustering metrics on Accidents after dimensionality reduction applied (Top: Dunn, DBI, CHI, Silhouette; Left: Silhouette plot; Right: BIC curve)

Neural Networks

The previously tuned network aims to learn non-linear mappings using layered perceptrons. It evaluates the effectiveness of the prior techniques by running on the original full, reduced, and cluster-derived datasets and using a comprehensive set of performance metrics. The held-out neural network analysis is only run on Accidents. It consists of two 512-node hidden layers applying the ReLu activation function. Preliminary testing defined `SGD` from PyTorch as the optimal model [10], incorporating momentum to avoid overfitting the noise. The hyperparameters were set as follows: `max_updates=250`, `max_iterations=5`, `learning_rate=0.05`, `betas=(0.9, 0.999)`, `weight_decay=1e-2`, `dropout_p=0.25`, `l_threshold=0.48`, `'label_smoothing_alpha=0.0`.

Observations and Analysis

Clustering Raw Data

K-means shows an impressively efficient convergence when run on the Hotels set, selecting five clusters as its optimal configuration. A moderate Silhouette score indicates medium cluster cohesion and suggests some overlap of feature patterns. This may be due to the large class imbalance, which assigns moderate patterns to the larger class. It scores higher than GMM across all metrics (a lower DBI is better), suggesting that hard cluster assignments manage class imbalance better, in agreement with our hypothesis. K-Means achieving the highest Dunn Index implies a favorable internal spread compared with measurable class overlap through probabilistic evaluations. GMM scoring a concerning Adjusted Rand Index(ARI) across five runs with seven clusters shows strong variability, confirming that hard, centroid-based assignments reduce sensitivity to clustering imbalanced datasets.

Both algorithms run on Accidents selected the fewest clusters available in the specified range, suggesting that a more simplistic approach, such as binary classification, scales well despite the distracting noise. Regardless, modest Silhouette scores, in conjunction with a higher DBI, imply that the overlap is most likely caused by the continuous duration target. Though practically perfect stability demonstrates that a dual-cluster configuration should regularly outperform other cluster counts, it supports our claim that hard clustering excels in large, noisy sets by consistently partitioning subtle but overarching patterns.

The higher scores for Hotels suggest that clustering manages imbalanced class separation better than mitigating noise. A significantly longer evaluation duration for GMM on Accidents implies that simplicity does not reduce overall computation.

Dimensionality on Raw Data

Principle Component Analysis: With seven components selected for the Hotels set, PCA achieved a moderately high cumulative explained variance, capturing a majority of the dataset structure while splitting the fourteen original features in half. After the top two components, the explained variance ratios demonstrate a steep drop and suggest the first few features that dominate the variance. PCA prioritizes features with higher variance, likely including client lead time or month, while muting noise from irrelevant attributes.

On the other hand, the explained variance ratios suggest that features in Accidents display a relatively equal global influence, with loadings demonstrating a gradual decrease in ranking among them. Excessive feature count with few dominating patterns aligns with its noisy structure, but PCA excels at locating which ones are most relevant and performs better than its competitors. The reconstruction error values further imply Hotels' imbalanced nature, losing minimal structure from dropping weak attributes, yet leaving a devastating impact when stripping the Accidents set of many of its uncorrelated signals.

Independent Component Analysis: ICA selected six components for the Hotels set, with a higher mean and absolute kurtosis indicating a strong non-Gaussian structure across the extracted components. Higher values among the most dominant components suggest that sparse signals are successfully isolated despite the imbalance. The emphasis on statistical separation exposes hidden drivers, such as seasonal trends or client family structures, supporting our hypothesis that the Hotels dataset contains few independent features with strong signals, leading to high imbalances. The results repeatedly showed high scores for Hotels running ICA with five to seven clusters.

The larger set derived seven components with ICA but showed weak independence among them, indicated by the low kurtosis score. It required significantly more execution time compared to the other algorithms, demonstrating difficulties in converging in noisy settings. Large datasets dominated by noisy patterns contain few independent signals for ICA to exploit. As predicted, it struggled to define individual components despite its extended resource consumption.

TABLE II
CLUSTERING RESULTS ON ORIGINAL DATA (STEP 1)

Dataset	Algorithm	k/n	Sil.	CHI	DBI	Dunn	ARI	Wall Time (s)
Hotels	K-Means	5	0.2237	14,569	1.5129	0.4076	0.7010	41.97
Hotels	GMM	7	0.1771	11,498	1.7228	0.3555	0.5573	77.14
Accidents	K-Means	2	0.1809	171,381	2.2427	0.3249	0.9987	99.08
Accidents	GMM	2	0.1743	170,261	2.2994	0.3174	1.0000	116.24

Random Projection: Pairwise distances are preserved with RP by multiplying the original high-dimensional data by randomly generated matrices. The entries are stored in a dense Gaussian Random matrix sampled independently from a standard normal distribution. RP was measured as the most computationally inexpensive algorithm, sacrificing accuracy for speed. It displayed notable structural distortion with poor reconstruction scores, which proved to be worse with Accidents. Due to its random nature, RP performs better with features that are more evenly distributed, scoring well on the lower-dimensional Hotels set, while the Accidents dataset amplifies loss with less redundancy.

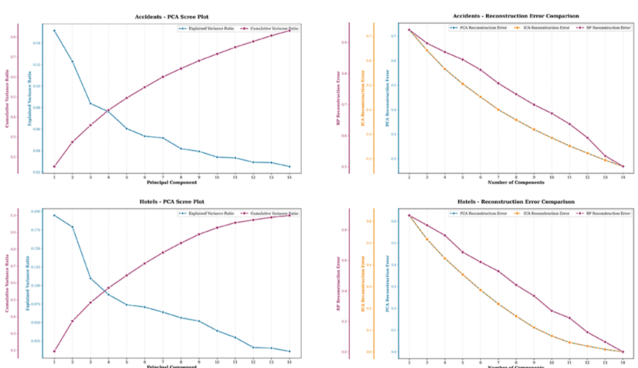

Fig. 7. Left: PCA Scree Plot(Cumulative and Explained variance); Right: Reconstruction error across all DR algorithms; Top: Accidents; Bottom: Hotels

CLUSTERING IN DIMENSIONALITY-REDUCED DATA

After reducing the number of features within each dataset and determining the optimal number of components for each algorithm, the sets are re-clustered using K-Means and GMM once again. Their results evaluate the effectiveness of dimensionality reduction and whether cluster quality, stability, and separation are preserved. This step strives to assess how removing noisy or redundant dimensions affects the meaningful signals dominating the clustered structure. K-Means clustering proves to maintain robust predictions with hard assignments, while GMM trades its stability for speed and suffers from fragile covariance in low dimensional settings.

Hotels: ICA reflected lower and more defined intra-cluster distances. GMM showed increased stability due to components emphasizing independent booking signals with ICA. It scored the highest ARI and kurtosis among the others, supporting the ICA-Hotels claim that ICA would isolate behavioral patterns that encourage cleaner class separation.

The Hotels dataset substantially improved from DR, with both component analysis techniques confirming that correlated patterns in client bookings can be easily separated due to minimal noise. PCA displayed tighter clusters, indicating further centroid separation and lower dominance by the larger class. DBI also decreased and improved overall clustering compactness.

RP scored the highest reconstruction error, losing the most structure from the original dataset. Outperforming the others, silhouette scores and improved cluster separation were observed after removing noise. RP trades structure for more efficient separation.

Higher Silhouette and lower DBI scores imply that cluster separation improved almost universally after applying dimensionality reduction algorithms. Stability increased overall, with significant improvements in K-Means runs, supporting our claim that hard data assignments are more robust and lead to clearer results in lower-dimensional spaces. GMM also substantially improved speeds after removing many covariance parameters, making estimation maximization far more efficient.

Accidents

Dimensionality Reduction condenses 28 engineered features into five or seven, risking substantial structural distortion it encourages naturally overarching separative factors and redefines many noisy features based on variance. Fewer, cleaner dimensions retain primary groupings while filtering out distracting patterns. The stricter assignments delivered by K-Means generate more gains than GMM, aligning with our claim that stronger datum decisions are better for noisier datasets. Overall, fewer features sped up the clustering algorithms by just under fifty percent.

PCA delivers the dominant variance patterns within the noisy dataset and maintains data spread across components. While probabilistic assignments offer slight separational improvements, stability crashes after removing the minor patterns. Meanwhile, K-Means leverages this to produce sharper and more distinct cluster centroids than prior to feature reduction. PCA joins many covariant features to deliver a few cumulative signals, unmasking correlated and meaningful patterns. It excels in sets with high variance and overall confirms its anticipated ability to successfully split noisy data with hard point decisions.

Results for ICA are similar to those reported from PCA, suggesting that collinear features overshadow independence. Scoring a low kurtosis still implies that the set has weak non-Gaussian signals after dimensionality reduction, limiting the potential for unique isolation. K-means clustering shows robust group separation without fracturing the structure, while GMM demonstrates clear stability but underperforms. ICA attempts to distinguish overlapping features and separate the strongest "voices" within the Accidents data, but it proves to be far too noisy, resulting in signal distortion.

Random averaging smooths out the noise and reveals small groupings hidden in higher dimensions. Clustering with RP shifts cluster counts from two to three, allowing for slightly more generalization while explaining its lower stability. GMM adds to instability with soft clustering, while K-Means benefits from preserving distances to enhance separation metrics. New perspectives emerge from random generations. However, they add some level of distortion and naturally vary between runs.

Neural Networks in Dimensionality-Reduced Data

A control neural network model is first trained on the Accidents data achieving a test loss of 0.6371 mean squared error. It uses previously tuned hyperparameters and is measured as a baseline for subsequent tests. Then the same network is retrained on the reduced data from each DR algorithm, supplying clear metrics that expose the true performance of dimensionality reduction in an applicable setting and whether learning quality improves.

PCA delivers moderate performance loss after removing redundant correlations, as it is unable to focus on individual signals in the noise. A higher MSE suggests insignificant variance, discarding subtle signals within the collinear noise. Elusive features that offer diminishing covariant gains, such as temperature and humidity, are pruned for greater efficiency.

ICA outperforms the others but remains worse than the baseline, with a six percent increase in mean square error results. The network is provided with the strongest signals to train from, again with more meaningful weight assignments between layers. However, it loses too much information after reduction. ICA helps expose and isolate non-Gaussian tails despite the loss.

RP shows worse efficiency and blurs fine structural information, demonstrating a difficulty in retaining bulk distances, introducing unintended artifacts, and likely creating a blurry representation from a clear image. Its loss plateaus at a higher value due to the network fitting artifacts generated by randomness, which agrees with our claim.

Overall, noise was muted across all three algorithms to expose the pattern structure within the Accidents dataset. With heavily distracting noise and equally influential features, dedicated component derivation using ICA and PCA confirms clear advantages when compared to an arbitrary guess and check methodology. Random Projection struggled on the Accidents set, while ICA and PCA were measured as worse than the baseline but demonstrated meaningful boosts in feature reduction efficiency. While this rejects the expected outcome given our hypothesis, the fundamental comparisons between the dimensionality reduction results align with our reasoning. When performing DR on a dataset, information removal is inevitable, leading to a loss in accuracy, but reduced parameter counts offer potential complexity advantages [4].

Neural Networks on Cluster-Derived Reduced Data

The final experimental step involves appending the reduced Accidents dataset with the cluster components derived from the previous clustering step. Rather than relying exclusively on a few broad clusters, the exposed unsupervised structures are used in conjunction with dimensionality reduction to generate a list of optimally meaningful and relevant features.

Compared to the base models, adding GMM-derived components to the feature set yielded mixed results among its counterparts but significantly poorer metrics compared to the baseline. GMM often outperforms K-Means, implying a diluted but still apparent advantage over hard assignments. EM uses probabilistic memberships to prevent rigid overfitting to noise. Although EM displays slightly higher prediction complexity by generating more clusters than its counterpart, it supports the hypothesis that soft membership smooths gradients within noisy data and should outperform stricter boundaries that are unable to separate overlaps [3].

Incorporating GMM with PCA proves to be a strong pipeline configuration, combining wide variance with probabilistic cluster assignments, similar to a high-contrast map with zoned districts. It demonstrates a 3.7 percent performance increase over PCA alone, but it is significantly worse than the baseline, while ICA scores have dropped sharply. PCA delivered an improved MSE but ICA returned a larger metric. Enforced feature independence from ICA appears to dilute beneficial overlapping patterns and lead to exaggerated overfitting, especially with hard K-Means assignments. Non-Gaussian source separation helps isolate dataset signals but risks losing relational information between features.

When RP is combined with GMM it surprisingly achieves the best dual-configuration results, most likely using the derived components as stabilizers across the randomized matrix. Even though RP was the worst stand-alone dimensionality reduction method, including soft cluster boundaries introduces benefits that smooth out the chaotic gradient generated by random projections. The fundamental idea that focusing on configuration and methodology over appending more machine learning techniques echoes our claim.

Dataset reductions offer minor speed and performance when optimal configurations are generated with complementary clustering methods. Though appending derived components to the reduced feature space offers additional optimization, the deleted information seems to be critical in navigating the feature space. This segment challenged our stated expectation of seeing universal performance advantages from synergizing unsupervised learning algorithms, clearly displaying variable results. But following our review of each technique's strengths and weaknesses, successful combinations evidently emerge.

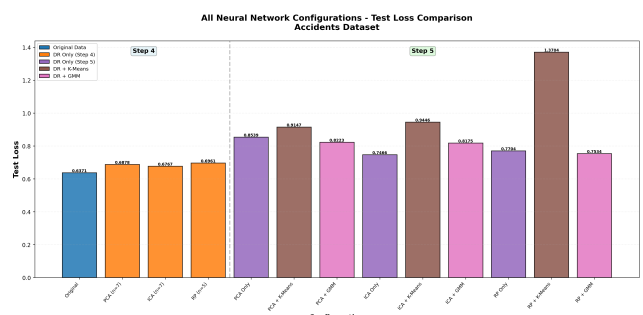

Fig. 8. Measured Test Loss of all Neural Network configurations

Conclusion

Optimized Pipeline

Key Findings

Clustering supported our hypothesis, revealing imbalanced structures in Hotels that scored higher in K-Means than GMM. K-Means incorporates hard decisions and avoids probabilistic uncertainty, thereby reducing overlap between classes. Dimensionality reduction exposed dominant features within each dataset, further demonstrating PCA success due to its ability to combat even features, capturing hidden gradient structures via the top eigenvalues across collinear noise. ICA has been shown to be suitable for the Hotels set by isolating non-Gaussian signals and

TABLE V
NEURAL NETWORK PERFORMANCE ON ACCIDENTS DATASET: DR-ONLY VS. DR + CLUSTERING-DERIVED FEATURES

Configuration	DR Dim	Clusters Added	Total Dim	Test MSE	Train MSE	Time (s)	N Params	%△ vs DR-only	%△ Orig.
PCA only	7	0	7	0.8539	0.8520	29.44	35,073	—	+34.0%
PCA + K-Means	7	4	11	0.9147	0.9116	29.95	—	+7.1%	+43.5%
PCA + GMM	7	18	25	**0.8223**	0.8191	29.21	—	–3.7%	+29.0%
ICA only	7	0	7	0.7466	0.7442	30.11	35,073	—	+17.2%
ICA + K-Means	7	12	19	0.9446	0.9399	28.91	38,145	+26.5%	+48.2%
ICA + GMM	7	9	16	0.8175	0.8158	30.48	37,377	+9.5%	+28.3%
RP only	5	0	5	0.7704	0.7693	28.77	34,561	—	+20.9%
RP + K-Means	5	6	11	1.3704	1.3673	28.47	36,097	+77.9%	+115.1%
RP + GMM	5	9	14	0.7534	0.7523	28.55	36,865	–2.2%	+18.2%

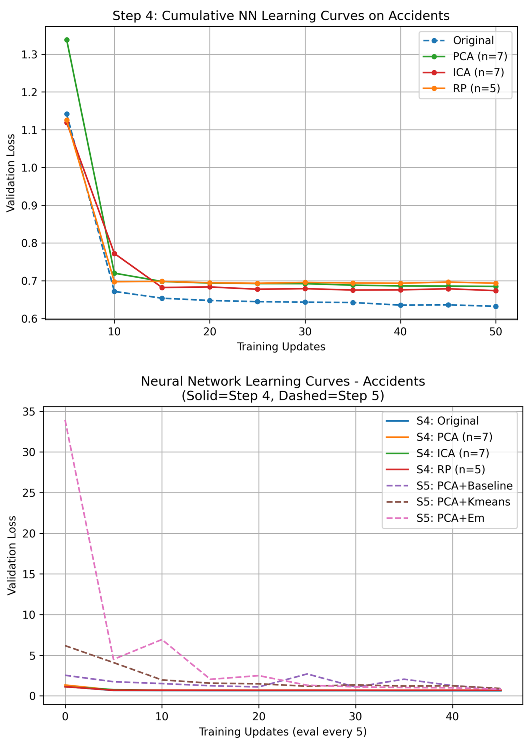

Fig. 9. Neural Network Learning curve on Accidents (Top: before DR; Bottom: after DR)

separating independent patterns [1]. Running the configurations through neural networks highlighted the hidden nuances of manipulating data. Due to immense information removal, the clustered components ignored critical signals and performed worse than the would have without.

TABLE VI
BEST PERFORMING PIPELINES AND KEY TAKEAWAYS

Dataset	Best Pipeline	Key Metric
Hotels	ICA + K-Means	Silhouette 0.290 ARI 0.996
Accidents	RP + GMM → NN	Test MSE 0.753 Stability ARI 0.785

Limitations and Future Work

This intensive study was an accumulation of a multitude of thorough machine learning techniques and their implementations. The complex nature of the multi-algorithm pipeline scales rapidly, compounding run times with each module interaction. Complexity remained a bottleneck for quick production and analysis. Future development will involve further abstracting logic behind a stronger object-oriented design, modularizing components for more efficient programming and execution.

ARTIFICIAL INTELLIGENCE USE STATEMENT

Artificial intelligence was used in the planning, coding, debugging, formatting, and error checking stages of this project. GitHub's Copilot [11], leveraging OpenAI's ChatGPT [12], Anthropic's Sonnet [13], and xAI's Grok [14] were used to brainstorm report and software designs, generate functions (i.e. helpers, plotters, readme.md), and bugfix code. Microsoft Word's embedded tool [15] helped fix spelling and grammar, and Google's Gemini [16] made internet searches and documentation referencing more efficient.

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